Google AI Studio Gemini API Key Usage and Quota Guide

Usage quotas and rate limits for a Gemini API Key issued by Google AI Studio depend on whether you are using the **Free Tier** or the **Paid Tier (Pay-as-you-go)**.

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1. Free Tier

The default tier available at no cost without registering a credit card. It is suitable for prototyping, personal development, and lightweight testing.

- **Billing**: Completely free (no credit card required)
- **Key Features**: Free access to select core models such as the Flash family.
- **Data Policy**: Prompts and responses submitted under the Free Tier **may be used by Google to improve its products (including AI model training)**, so avoid submitting sensitive personal information or confidential corporate data.
- **Rate Limits**:
  - **Requests Per Minute (RPM)**, **Tokens Per Minute (TPM)**, and **Requests Per Day (RPD)** limits apply per project and per model.
  - Depending on the model (e.g., Flash series), usage is typically restricted to several to a few dozen requests per minute (RPM) and up to a few hundred requests per day (RPD).
  - Exceeding the limit results in an **HTTP 429 (Too Many Requests)** error and blocks the request.

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2. Paid Tier (Pay-as-you-go)

Linking a **Billing Account** to your Google Cloud project upgrades your project to the Paid Tier.

- **Billing Model**: **Pay-as-you-go** based on the number of tokens used
- **Key Features**: Unrestricted access to all models, including high-performance models such as the Pro family.
- **Data Privacy**: Data on the Paid Tier is **not used to train Google models** (ensuring security and compliance).
- **Tiered Rate Limits**:
  - **Tier 1**: Activated immediately upon linking a billing account (significantly increases per-10-minute spending limits and daily quotas)
  - **Tier 2 and Tier 3**: Automatically upgraded as cumulative spend and account age meet thresholds, guaranteeing much higher throughput (RPM / TPM).

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3. Tier Comparison Summary

Category	Free Tier	Paid Tier (Pay-as-you-go)
Cost	Completely free (no card required)	Pay-as-you-go (based on token usage)
Available Models	Primarily Flash family	Supports Pro and all latest models
Data Usage	May be used to improve/train models	Not used for model training (privacy guaranteed)
Request Limits	A few to dozens RPM / hundreds RPD	High-volume throughput across Tiers 1–3
Best Suited For	Personal development, prototyping, learning/testing	Commercial services, production deployment, batch workloads

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4. How to Check Exact Real-Time Quotas

Because exact numbers may vary depending on Google policies or model versions, you can check the current limits applied to your API key as follows:

1. **Visit and sign in to [Google AI Studio]**
2. In the left navigation menu, go to **Dashboard** or **API keys**, then check **Usage and limits**
3. View your current **RPM** (Requests Per Minute), **TPM** (Tokens Per Minute), and **RPD** (Requests Per Day) limits and real-time usage for each model

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5. Environment Variable Setup (`.env`) and Vertex AI Integration Guide

In the `.env` file at the root of the labs, configure authentication by choosing either the **Google AI Studio API Key** method or the **Google Cloud Vertex AI** method.

① Using a Google AI Studio API Key (Default)

Register the API key issued from AI Studio directly:

```
GOOGLE_GENAI_USE_VERTEXAI=FALSE
GOOGLE_API_KEY=AIzaSy...
```

② Integrating with Vertex AI (Google Cloud)

Switch to Vertex AI if you want to use Google Cloud credits or run high-volume workflow/agent workloads without the per-minute rate limits (`429 Too Many Requests`) of a free API key.

1. **Prepare in the GCP Console**:
    - Create a project in the Google Cloud Console
    - Enable the **Vertex AI API** (`aiplatform.googleapis.com`)
2. **Sign in via `gcloud` CLI (Application Default Credentials)**:
    - Run the following command in your local terminal to complete account authentication:

```
gcloud auth application-default login
```

3. **Configure the `.env` File**:
    - Set `GOOGLE_GENAI_USE_VERTEXAI=TRUE` and enter your GCP project ID and location (comment out `GOOGLE_API_KEY`):

```
GOOGLE_GENAI_USE_VERTEXAI=TRUE
GOOGLE_CLOUD_PROJECT=your-gcp-project-id
GOOGLE_CLOUD_LOCATION=asia-northeast3   # Seoul region (or us-central1)
# GOOGLE_API_KEY= (commented out)
```

> 💡 **Tip**: Switching to Vertex AI allows you to reliably run the `google_search` tool integration in `part06_tools` , large-scale evaluations (`adk eval`), and multi-agent pipelines without worrying about quota limits.